

Analytical Solution of MSE – Questions and Answers

Q: What is the formula for the Mean Squared Error (L_MSE or L2 Loss)?

A: $L_MSE = (1/n) \sum (y_i - \hat{y}_i)^2$.

Q: Minimizing the MSE is equivalent to what type of statistical estimation, and under what noise assumption?

A: Minimizing MSE equals performing Maximum Likelihood Estimation under Gaussian noise.

Q: The example of $\hat{y} = \beta x$ involves a linear model with how many variables?

A: The model $\hat{y} = \beta x$ involves one independent variable.

Q: What is the general analytical approach used to find the optimal parameter (β) that minimizes the MSE?

A: The analytical approach involves differentiating the loss function and setting the derivative to zero.

Q: In the context of L2 loss, which term in $(y_i - \hat{y}_i)^2$ represents the predicted value?

A: The term $\hat{y}_i$ represents the predicted value.

Q: Write down the partial derivative of L2 with respect to β for the model $\hat{y} = \beta x$.

A: $\partial L / \partial \beta = -(2/n) \sum x_i (y_i - \beta x_i)$.

Q: By setting the partial derivative $\partial L / \partial \beta$ to zero, what is the resulting equation for β ?

A: Setting $\partial L / \partial \beta = 0$ gives $\sum x_i y_i - \beta \sum x_i^2 = 0$.

Q: What is the final analytical solution for the best-fit parameter β in the $\hat{y} = \beta x$ model?

A: The solution is $\beta = (\sum x_i y_i) / (\sum x_i^2)$.

Q: What do X and Y represent in the matrix form $\beta = (X^T X)^{-1} X^T Y$?

A: X is the matrix of inputs, and Y is the vector of outputs.

Q: The analytical solution for the best-fit parameter is referred to as the equation for the what?

A: It is referred to as the Normal Equation.

Q: What is the equation for the predicted value $\hat{y}(x)$ in the quadratic regression example?

A: $\hat{y}(x) = \beta_0 + \beta_1 x + \beta_2 x^2$.

Q: For the quadratic regression model, how many parameters (β values) must be solved for?

A: The quadratic regression has three parameters: β_0 , β_1 , β_2 .

Q: To find the optimal parameters β_0 , β_1 , and β_2 for the quadratic model, how many partial derivatives must be set to zero?

A: Three partial derivatives (one for each parameter) must be set to zero.

Q: Write down the simplified linear equation derived by setting $\partial MSE / \partial \beta = 0$.

A: $\sum y_i = n\beta_0 + \beta_1 \sum x_i + \beta_2 \sum x_i^2$.

Q: What is the partial derivative used to solve for β_1 in the quadratic model?

A: $\partial \text{MSE} / \partial \beta_1 = -(2/n) \sum x_i (y_i - \beta_0 - \beta_1 x_i - \beta_2 x_i^2)$.

Q: Write down the simplified linear equation derived by setting $\partial \text{MSE} / \partial \beta_1 = 0$.

A: $\sum x_i y_i = \beta_0 \sum x_i + \beta_1 \sum x_i^2 + \beta_2 \sum x_i^3$.

Q: What is the simplified linear equation derived by setting $\partial \text{MSE} / \partial \beta_2 = 0$?

A: $\sum x_i^2 y_i = \beta_0 \sum x_i^2 + \beta_1 \sum x_i^3 + \beta_2 \sum x_i^4$.

Q: The three equations for β_0 , β_1 , and β_2 form what type of system?

A: They form a system of linear equations.

Q: What is the term in the first row, third column of the table representing the system?

A: The first row, third column term is $\sum x_i^2$.

Q: What is the term in the second row, third column of the table?

A: The second row, third column term is $\sum x_i^3$.

Q: The proposed analytical approach using MSE can be extended to models with a greater number of what?

A: The analytical method can be extended to models with more parameters.

Q: The analytical approach can be readily extended to more what kind of models?

A: It can be extended to more complex polynomial or linear models.

Q: What is the name of the numerical optimization algorithm typically used for optimizing parameters?

A: The typical algorithm used is Gradient Descent.

Q: When parameters are optimized numerically, what are the common variations of the algorithm used?

A: Common variations include Stochastic and Mini-Batch Gradient Descent.

Q: When a model is very complex, is the optimization of parameters typically done analytically or numerically?

A: For complex models, parameter optimization is performed numerically.

Q: For the simple linear model $y = \beta_0 + \beta_1 x$, what term is associated with β_1 in $\sum (x_i y_i - \beta_0 x_i - \beta_1 x_i^2)$?

A: The term associated with β_1 is $-\beta_1 \sum x_i^2$.

Q: In the quadratic regression problem, the term $\sum x_i^3$ appears on the left-hand side of which parameter equation?

A: The term $\sum x_i^3$ is associated with β_1 .

Q: In the table of equations, which summation appears in the diagonal element corresponding to β_j ?

A: The diagonal element for β_j is $\sum x_{ij}^2$.

Q: Based on the table, what is the right-hand side term for the equation derived from $\partial \text{MSE} / \partial \beta_j = 0$?

A: The right-hand side term is $\sum y_{ij}$.

Q: When solving for $\beta_j = (\sum x_{ij}^2)^{-1} (\sum x_{ij} y_{ij})$, what must be assumed about $\sum x_{ij}^2$?

A: It must be assumed that $\sum x_{ij}^2 \neq 0$ so that its inverse exists.